Diploma Semester 1 • Basic Science • Practical Handbook

Applied Chemistry Lab

This is a screen-filling practical handbook for Course Code 1007. It is written as a lab companion: theory before the bench, procedure at the bench, calculation after the bench, and viva preparation before evaluation. It follows the official syllabus structure but expands each experiment into a usable student manual.

1007
Course Code

30
Periods / Semester

2
Practical periods / week

1
Credit

4
Course Outcomes

6+
Minimum Experiments

Official Syllabus Map

The lab is not a theory subject. Marks come from clean handling, accurate observation, correct calculation, disciplined record writing and clear viva answers.

Course Objectives

- Supplement lecture knowledge through first-hand handling of apparatus, reagents and analytical processes.
- Develop scientific temper and apply basic chemistry principles to engineering problems.
- Train students to report practical results in a standard, reproducible format.

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Course Outcomes

CO	Outcome	Hours	Level
CO1	Quantitatively analyse solutions accurately.	8	Applying
CO2	Standardise EDTA and analyse hardness of water.	4	Applying
CO3	Determine pH of solutions using different techniques.	4	Applying
CO4	Apply electrochemistry in quantitative analysis.	6	Applying

Module Flow

CO1 • 8 h

Volumetric Analysis

Oxalic acid, HCl standardisation, NaOH estimation and KOH strength.

CO2 • 4 h

Complexometry

EDTA standardisation and total hardness of water.

CO3 • 4 h

pH Measurement

pH meter and universal indicator paper.

CO4 • 6 h

Electrochemistry

Conductivity, Faraday's first law and Daniel cell EMF.

CO–PO Mapping

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3			3		
CO3	3						
CO4	3						

3 = Strongly mapped. For lab work, PO1 is dominant because the student applies science and basic engineering knowledge directly through measurement.

Laboratory Safety and Discipline

Most wrong results in chemistry lab are not caused by difficult theory. They are caused by careless washing, parallax error, air bubbles, wrong indicator, and bad record writing.

Personal Safety

- Wear lab coat, shoes and eye protection where instructed.
- Never pipette by mouth. Use a pipette filler.

- Wash skin immediately after acid, alkali or salt contact.
- Keep bags, food and water bottles away from the bench.

Accuracy Rules

- Read burette at eye level from the lower meniscus.
- Remove air bubbles from burette tip before starting.
- Do rough titration first, then concordant trials.
- Concordant titre usually means readings within 0.1 mL.

Do Not Do

- Do not pour excess reagent back into reagent bottle.
- Do not write guessed readings.
- Do not heat closed containers.
- Do not touch electrode surfaces unnecessarily.

Experiment Handbook

Each experiment is expanded with aim, theory, apparatus, procedure, observation, calculation, result, precautions and viva points.

M1.01 • CO1 • 2 h

Preparation of Standard Oxalic Acid Solution

A standard solution has accurately known concentration. Oxalic acid is used because it is a primary standard: comparatively pure, stable and easy to weigh.

Theory

For hydrated oxalic acid $H_2C_2O_4 \cdot 2H_2O$, equivalent weight = molecular mass / basicity = 126 / 2 = 63 g/eq. To prepare 250 mL of N/10 solution, required mass = Normality × Equivalent weight × Volume in litre = 0.1 × 63 × 0.25 = 1.575 g.

Standard solution □□□□□□□□□□□□□□□□□ weighing □□□□□□□□□□□□□□□□□. Volumetric flask mark □□□ □□□□□□ make up □□□□□□□; mark □□□□□□□ concentration □□□□□□□.

Procedure

1. Weigh the calculated mass of oxalic acid using a clean weighing bottle.
2. Transfer it into a beaker and dissolve in a small quantity of distilled water.
3. Transfer the solution into a 250 mL standard flask through a funnel.
4. Wash the beaker and funnel into the flask to avoid loss of solute.

5. Make up to the mark with distilled water, stopper and shake well.

Observation / Record

Item	Value
Required normality	N/10
Volume	250 mL
Equivalent weight	63 g/eq
Mass required	1.575 g

Result

250 mL of approximately/accurately N/10 oxalic acid solution is prepared.

Viva

Why oxalic acid? It is suitable as a primary standard because it can be weighed directly to prepare known concentration.

M1.02 • CO1 • 2 h

Standardisation of HCl using Standard Sodium Carbonate

The exact strength of HCl is found by titrating it against a standard sodium carbonate solution using methyl orange indicator.

Theory

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$. Sodium carbonate is a weak base and HCl is a strong acid. Methyl orange gives yellow in alkaline medium and reddish-orange at the end point.

Procedure

1. Rinse and fill burette with given HCl.
2. Pipette 20 mL standard sodium carbonate into a conical flask.
3. Add 2–3 drops methyl orange.
4. Titrate until yellow changes to pale orange.
5. Repeat to get concordant readings and calculate normality of HCl.

Calculation

Use N1V1 = N2V2. If sodium carbonate is N1 and volume is V1, and HCl titre is V2, then NHCl = NNa2CO3 × VNa2CO3 / VHCl.

Precaution

Do not use phenolphthalein for final complete neutralisation here; methyl orange is the suitable indicator for strong acid versus weak base titration.

M1.03 • CO1 • 2 h

Estimation of Sodium Hydroxide using Standard HCl

The strength of NaOH is estimated by neutralisation with already standardised HCl.

Theory

NaOH + HCl → NaCl + H2O. This is a strong acid–strong base titration. Phenolphthalein is commonly used; it is pink in alkali and colourless in acid.

Procedure

1. Fill burette with standard HCl and note initial reading.
2. Pipette 20 mL NaOH solution into conical flask.
3. Add phenolphthalein indicator.
4. Titrate till pink colour just disappears.
5. Calculate normality and strength of NaOH.

Formula

NNaOH × VNaOH = NHCl × VHCl. Strength in g/L = Normality × Equivalent weight. Equivalent weight of NaOH = 40.

Common Mistake

Overshooting the endpoint gives higher titre and wrong lower calculated strength of NaOH.

M1.04 • CO1 • 2 h

Strength of KOH using Standard Oxalic Acid

KOH is titrated against standard oxalic acid to determine its normality and strength.

Theory

$H_2C_2O_4 + 2KOH \rightarrow K_2C_2O_4 + 2H_2O$. KOH equivalent weight is 56.1 g/eq. Phenolphthalein is suitable because the endpoint is observed by disappearance of pink colour.

Procedure

1. Fill burette with standard oxalic acid.
2. Pipette 20 mL KOH into conical flask.
3. Add phenolphthalein and titrate to colourless endpoint.
4. Repeat for concordant titre.
5. Calculate normality and strength of KOH in g/L.

Result Format

Strength of given KOH = ______ g/L.

Malayalam Note

KOH concentration □□□□□□ known oxalic acid □□□□□□□□□□□□□□. Endpoint light pink □□□□□□ colourless □□□□□□ point □□□.

M2.01 • CO2 • 2 h

Standardisation of EDTA using $ZnSO_4$

EDTA forms stable 1:1 complexes with metal ions. It must be standardised before using it for hardness estimation.

Theory

In complexometric titration, EDTA reacts with Zn^{2+} in 1:1 molar ratio. Buffer maintains pH near 10. Eriochrome Black T indicates the endpoint by colour change from wine red to blue.

Procedure

1. Pipette standard $ZnSO_4$ solution into conical flask.

2. Add buffer solution and EBT indicator.
3. Titrate with EDTA until wine red changes to clear blue.
4. Repeat and calculate molarity/normality of EDTA.

Key Point

EDTA titrations depend strongly on pH. Without buffer, endpoint becomes poor and result is unreliable.

Viva

Why pH 10? The metal-EDTA complex and indicator colour change are reliable near pH 10.

M2.02 • CO2 • 2 h

Total Hardness of Water using Standard EDTA

Water hardness is due mainly to Ca^{2+} and Mg^{2+} ions. It is reported as mg/L or ppm as $CaCO_3$.

Theory

Ca^{2+} and Mg^{2+} form complexes with EDTA. At endpoint, EBT changes from wine red to blue. Hardness as $CaCO_3$ = (V × M × 100000) / sample volume in mL, where V is EDTA volume and M is EDTA molarity.

Procedure

1. Pipette measured water sample into conical flask.
2. Add buffer solution and EBT indicator.
3. Titrate with standard EDTA to blue endpoint.
4. Record concordant readings.
5. Calculate total hardness as ppm $CaCO_3$.

Engineering Use

Hard water causes scaling in boilers, heaters, cooling systems and plumbing. For maintenance engineers, hardness is not just chemistry; it affects service life and efficiency.

Malayalam Note

Hardness □□□□□□□□□□ pipe, heater, boiler □□□□□□□□□ scale □□□□□□□□□□. □□□□□□□□□ water analysis engineering maintenance-□□□ □□□□□□□□□□.

M3.01 • CO3 • 2 h

Determination of pH using pH Meter

A pH meter gives instrumental measurement of hydrogen ion activity. Correct calibration is more important than pressing the reading button.

Theory

pH = -log10[H+]. pH below 7 indicates acidic solution, pH 7 neutral and pH above 7 alkaline. The glass electrode develops a potential depending on hydrogen ion activity.

Procedure

1. Switch on pH meter and allow it to stabilise.
2. Calibrate using standard buffer solutions such as pH 4, 7 and 9/10.
3. Rinse electrode with distilled water and blot gently.
4. Immerse electrode in sample without touching bottom of beaker.
5. Wait for stable reading and record pH.

Do Not

Do not rub the glass electrode. Rubbing creates static charge and may damage the sensitive membrane.

Result

pH of the given solution = ______.

M3.02 • CO3 • 2 h

pH using Universal Indicator / pH Paper

This is a quick approximate pH method using colour comparison.

Theory

Universal indicator is a mixture of indicators that shows different colours for different pH ranges. It is less accurate than pH meter but useful for quick identification.

Procedure

1. Place a drop of sample on pH paper or dip the strip briefly.
2. Compare the colour immediately with standard chart.
3. Record approximate pH and nature of solution.
4. Repeat for different given samples.

Comparison

pH meter: accurate, needs calibration. pH paper: quick, cheap, approximate.

M4.01 • CO4 • 2 h

Conductivity of Water using Conductometer

Conductivity indicates the ionic content of water. More dissolved ions usually means higher conductivity.

Theory

Conductance is reciprocal of resistance. Specific conductance depends on cell constant and measured conductance. Conductivity is useful for water quality monitoring.

Procedure

1. Calibrate conductometer using standard KCl solution if instructed.
2. Rinse conductivity cell with distilled water and then sample.
3. Immerse cell fully in water sample without air bubbles.
4. Record conductivity after reading stabilises.
5. Clean cell after use.

Industrial Link

Conductivity is used in boiler water, cooling water, battery water and process water checks.

Malayalam Note

Water-□ dissolved salts □□□□□□□□□□□□□□ conductivity □□□□□. Reading stable □□□□□□□□ □□□ □□□□□□□□□□□□□□.

M4.02 • CO4 • 2 h

Verification of Faraday's First Law of Electrolysis

Mass deposited or liberated during electrolysis is directly proportional to quantity of electricity passed.

Theory

Faraday's first law: $m \propto Q$, and $Q = It$. Therefore, $m = ZIt$ where Z is electrochemical equivalent. In copper sulphate solution using copper electrodes, copper gets deposited at cathode.

Procedure

1. Clean, dry and weigh the cathode.
2. Set copper electrodes in copper sulphate solution.
3. Pass steady current for a known time.
4. Remove cathode, wash, dry and weigh again.
5. Compare mass deposited with quantity of electricity.

Observation

Reading	Value
Initial mass of cathode	____ g
Final mass of cathode	____ g
Current	____ A
Time	____ s

Formula

m = final mass − initial mass; $Q = It$.

M4.03 • CO4 • 2 h

Measurement of EMF using Daniel Cell

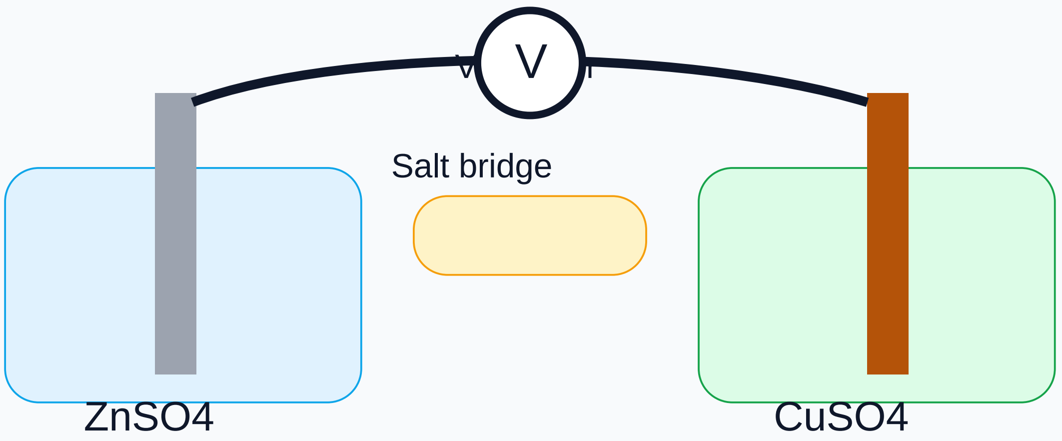
A Daniel cell converts chemical energy into electrical energy using zinc and copper half-cells connected by a salt bridge.

Theory

Cell notation: Zn | $ZnSO_4$ || $CuSO_4$ | Cu. Zinc acts as anode and copper as cathode. Electrons flow through the external circuit from zinc to copper.

Procedure

1. Set zinc electrode in zinc sulphate solution and copper electrode in copper sulphate solution.
2. Connect both half-cells using salt bridge.
3. Connect electrodes to voltmeter with correct polarity.
4. Record EMF after stable reading.
5. Repeat if concentration changes are given.



Result

EMF of Daniel cell = ______ V.

Formula Bank and Calculation Rules

Use these formulas carefully. Wrong units are the fastest way to lose marks in lab record and viva.

N1V1 = N2V2

Volumetric relation

For acid-base titration when reacting equivalents are matched.

Strength = Normality × Eq. wt.

Strength in g/L

Use equivalent weight of the substance being estimated.

Mass = N × Eq.wt × V(L)

Preparation

For preparing standard solution in a volumetric flask.

Hardness = VEDTA × M × 100000 / Vs

ppm as CaCO3

Vs is sample volume in mL.

pH = -log10[H+]

pH definition

Lower pH means higher acidity.

Q = I × t

Quantity of electricity

I in ampere, t in seconds.

m = ZIt

Faraday's first law

Mass deposited is proportional to charge passed.

Conductivity = Cell constant × Conductance

Conductometer

Check instrument units before writing result.

Record Writing Format

Use this order. Do not make your record look like random class notes.

Standard Practical Record Order

1. Date and experiment number
2. Aim
3. Apparatus and chemicals
4. Principle / theory
5. Procedure in past tense
6. Observation table
7. Calculation
8. Result
9. Precautions
10. Viva questions

Observation Table Discipline

- Never overwrite burette readings. Strike once and rewrite if correction is allowed.
- Use units in table headings, not repeated in every cell.
- Write initial and final readings clearly.
- Use at least one rough and two concordant readings.
- Result must match the calculation, not your expectation.

Sample Titration Table

Trial	Initial burette reading (mL)	Final burette reading (mL)	Titre value (mL)	Remarks
Rough				Approximate
1				Accurate
2				Concordant
3				If needed

Viva and Quick Revision

These questions cover the minimum expected oral preparation for the lab.

What is a standard solution?

A solution whose concentration is accurately known.

What is primary standard?

A pure, stable substance that can be weighed directly to prepare a standard solution.

Why remove air bubble from burette tip?

It gives wrong volume reading and therefore wrong titre value.

Why use indicator?

To show the completion of reaction by a visible colour change.

Endpoint vs equivalence point?

Equivalence point is theoretical completion; endpoint is observed colour change.

Why use buffer in EDTA titration?

To maintain pH around 10 for sharp endpoint and stable complex formation.

What causes hardness of water?

Mainly calcium and magnesium salts.

Unit of hardness?

Commonly ppm or mg/L as $CaCO_3$.

What is pH?

Negative logarithm of hydrogen ion concentration/activity.

Why calibrate pH meter?

To make the meter reading reliable against known buffer standards.

What is conductance?

Reciprocal of resistance.

State Faraday's first law.

Mass deposited during electrolysis is directly proportional to the quantity of electricity passed.

What is Daniel cell?

An electrochemical cell using zinc and copper half-cells connected by salt bridge.

Function of salt bridge?

It completes the internal circuit and maintains electrical neutrality.

Most common titration error?

Overshooting endpoint, parallax error, unwashed apparatus and air bubble in burette tip.

Model Practical Examination Pack

Use this as a preparation format for CIA/ESE. Actual experiment allotment depends on the institution/faculty.

Part A: Spot / Viva Questions

1. Define normality and molarity.
2. Why is oxalic acid used as a primary standard?
3. What is the colour change of methyl orange in acid-base titration?
4. Why EDTA titration requires buffer?
5. What is temporary and permanent hardness?
6. What is the principle of pH meter?
7. What is cell constant?
8. State Faraday's first law of electrolysis.
9. What is the role of salt bridge in Daniel cell?
10. Write any two precautions in titration.

Part B: Practical Tasks

1. Prepare standard N/10 oxalic acid solution and write calculation.
2. Standardise the given HCl using sodium carbonate.
3. Estimate the strength of NaOH or KOH from the given solution.
4. Determine total hardness of the supplied water sample.
5. Measure pH of two given samples using pH meter and pH paper.
6. Determine conductivity of a water sample.
7. Verify Faraday's first law using copper electrodes.
8. Measure EMF of Daniel cell.

Answer Key / Expected Points

No.	Expected Answer
1	Normality is gram equivalents per litre; molarity is moles per litre.
2	Oxalic acid is stable and can be accurately weighed for preparing standard solution.
3	Methyl orange changes from yellow to orange/red in acidic endpoint.
4	Buffer maintains required pH for EDTA-metal complex and sharp EBT endpoint.
5	Temporary hardness is due to bicarbonates; permanent hardness is due to chlorides/sulphates of Ca and Mg.
6	pH meter measures electrode potential related to hydrogen ion activity.
7	Cell constant is distance between electrodes divided by area of electrodes.
8	Mass deposited is proportional to charge passed through electrolyte.
9	Salt bridge completes circuit and maintains ionic balance.
10	Use clean apparatus, avoid parallax, remove air bubbles, add indicator carefully, take concordant readings.

References and Online Practice

Use these for deeper reading and simulation before entering the lab.

Text / Reference

- NCERT Chemistry Class XI & XII, Part I and II.
- G. H. Hugar and A. N. Pathak, Applied Chemistry Laboratory Practices, NITTTR Chandigarh.
- Rajesh Agnihotri, Chemistry for Engineers, Wiley India.
- Jain & Jain, Engineering Chemistry, Dhanpat Rai and Sons.

Online Resources

- Virtual Labs IIT Bombay: https://vlabs.iitb.ac.in
- Amrita Virtual Lab: https://vlab.amrita.edu